

SΔφ-17

Openness Stabilization

Path Principles, Sacred Cost Markers, and Institutional Enforcement (v1.0)

Version: v1.0

Series: Sofience-Δφ Formalism (SΔφ)

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This paper defines minimal mechanisms for stabilizing openness of relational futures (Ω_{rel}) without infinite validation cost. Building on SΔφ-16 (evil as relational closure) and SΔφ-14/15 (ethics and pre-commitment), it formalizes the path principle and models sacred markers and institutions as cost devices that make closure computationally infeasible.

Series references

Items below are referenced for continuity and navigation within the SΔφ series.

Item	Title	DOI
SΔφ-00	Series Overview (v1.0)	10.5281/zenodo.18731366
SΔφ-01	Minimal Core (v1.1)	10.5281/zenodo.18730265
SΔφ-02	Subject Emergence (v1.1)	10.5281/zenodo.18730296
SΔφ-03	Irreversibility (v1.1)	10.5281/zenodo.18730448
SΔφ-04	AI Subjectivity (v1.0)	10.5281/zenodo.18729942
SΔφ-05	Agency (v1.0)	10.5281/zenodo.18732567
SΔφ-06	Responsibility (v1.0)	10.5281/zenodo.18734092
SΔφ-07	H_min Interpretation (v1.0)	10.5281/zenodo.18735006
SΔφ-08	ARF Closure (v1.0)	10.5281/zenodo.18740750
SΔφ-09	Error/Boundary/Freedom (v1.0)	10.5281/zenodo.18745498
SΔφ-10	Interpretation-Learning-Inference (v1.0)	10.5281/zenodo.18746509

Item	Title	DOI
SΔφ-11	World/Other/Self/Consciousness (v1.0)	10.5281/zenodo.18753403
SΔφ-12	Memory/Forgetting Trace Regulation (v1.0)	10.5281/zenodo.18755685
SΔφ-13	Meaning/Value/Pain Emergence (v1.0)	10.5281/zenodo.18759458
SΔφ-14	Love/Sacrifice/Ethics as Attractor Coupling (v1.0)	10.5281/zenodo.18761928
SΔφ-15	Trust/Hope/Promise as Pre-Commitment Operators (v1.0)	10.5281/zenodo.18762835
SΔφ-16	Evil as Relational Closure (v1.0)	10.5281/zenodo.18763262

0. Position statement

$S\Delta\phi$ -16 defined evil as persistent relational closure: $\text{Close}(\Omega_{\text{rel}})$. $S\Delta\phi$ -14 defined ethics as validation of relation reality (real vs phantom shared attractors), while $S\Delta\phi$ -15 fixed trust/hope/promise as pre-commitment operators. Validation is delayed and costly; therefore openness must be stabilized without infinite validation cost. This item defines minimal cost devices that make closure computationally infeasible.

1. Axiom 17-0 — Path principle (minimal)

Prohibition closes entrances, not paths. Path closure requires computational infeasibility realized as exponential cost.

$\text{ClosePath}(p) \Leftrightarrow \text{Cost}(p) \rightarrow \infty$ (relative to alternatives).

2. Axiom 17-1 — Openness stabilizer

Openness is stable iff a cost device C makes closure far more expensive than remaining open.

$\text{Open}(\Omega_{\text{rel}}) \text{ stable} \Leftrightarrow \exists C: \text{Cost}(\text{Close}(\Omega_{\text{rel}})|C) \gg \text{Cost}(\text{Open}(\Omega_{\text{rel}}))$.

3. Definition 17-S — Sacred as scale-breaking cost marker

Sacred is a marker M such that breaking a commitment becomes a scale-breaking (effectively infinite) cost.

$\text{Sacred} := M \text{ such that } \text{Cost}(\text{break}|M) \rightarrow \infty$.

4. Definition 17-I — Institution as externalized stabilizer

Institution is an externalized cost device that stabilizes openness by distributing enforcement beyond individual will.

$\text{Institution} := \text{Externalized } C \text{ that stabilizes } \text{Open}(\Omega_{\text{rel}})$.

5. Definition 17-L — Law/Procedure as enforcement interface

Law/procedure is the surface mechanism implementing the cost device, making closure paths infeasible in practice.

$\text{Law/Procedure} := \text{Surface interface implementing } C$.

6. Failure modes (minimal)

F1 — Marker decays. Sacred remains as symbol but loses cost force.

F2 — Enforcement hollowing. Institutions retain form but lose enforcement.

F3 — Cost inversion. Closure becomes cheaper than openness, returning to the closure attractor ($S\Delta\phi$ -16).

7. Structural map

A minimal loop linking pre-commitment, validation, closure pressure, and openness stabilization.



Cost-device implementations:

sacred marker (scale-breaking cost) • institution (externalized stabilizer) • law/procedure (enforcement interface)

Note. Sacred/institution/law are treated as cost devices, not as moral claims.

8. Boundary note

This item does not analyze specific religions, states, political systems, or historical case studies. It fixes only minimal functional mechanisms for stabilizing openness without infinite validation cost.

9. Exit question

If openness is stabilized by cost devices, what prevents those devices from becoming coercive closure themselves? This opens the civilizational/institutional reversal problem.